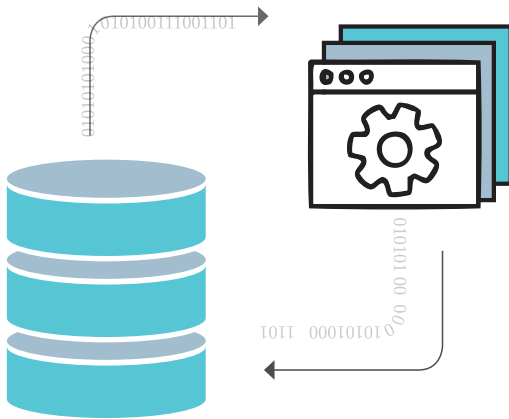


Transforming Data with R

In this fifth article in the 'R, Statistics and Machine Learning' series, we shall learn the various R functions that are available to combine, modify, select and apply functions on data.



We will be using R version 4.1.0 installed on Parabola GNU/Linux-libre (x86-64) for the code snippets given here.

```
$ R --version
R version 4.1.0 (2021-05-18) -- "Camp Pontanezen"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)
R is free software and comes with absolutely no warranty.
```

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Merge

The `paste()` function concatenates multiple character vectors into a single vector, as shown below:

```
> a <- c("1.", "2.", "3.", "4.")
> b <- c("Chapter: Introduction", "Chapter: Trees", "Chapter:
Graphs", "Chapter: Networks")
> paste(a, b)
[1] "1. Chapter: Introduction" "2. Chapter: Trees"
[3] "3. Chapter: Graphs"      "4. Chapter: Networks"
```

You can also pass a separator string to combine the character vectors. In the following example, we use the `'.'` (dot) to combine strings.

```
> a <- c("1", "2", "3", "4")
> b <- c("Chapter: Introduction", "Chapter: Trees", "Chapter:
Graphs", "Chapter: Networks")
> paste(a, b, sep=". ")
[1] "1. Chapter: Introduction" "2. Chapter: Trees"
[3] "3. Chapter: Graphs"      "4. Chapter: Networks"
```

Consider the `mtcars` data set available in R. You can add a column to the data frame or matrix, using the `cbind()` function, as shown below:

```
> head(mtcars)
mpg cyl disp hp drat wt qsec vs am gear carb
Mazda RX4 21.0 6 160 110 3.90 2.620 16.46 0 1 4 . 4
Mazda RX4 Wag 21.0 6 160 110 3.90 2.875 17.02 0 1 4 4
Datsun 710 22.8 4 108 93 3.85 2.320 18.61 1 1 4 1
Hornet 4 Drive 21.4 6 258 110 3.08 3.215 19.44 1 0 3 1
Hornet Sportabout 18.7 8 360 175 3.15 3.440 17.02 0 0 3 2
Valiant 18.1 6 225 105 2.76 3.460 20.22 1 0 3 1

> data <- head(mtcars)
> year <- c(1970, 1970, 1973, 1974, 1977, 1962)
> new_mtcars <- cbind(data, year)
> new_mtcars
mpg cyl disp hp drat wt qsec vs am gear carb year
Mazda RX4 21.0 6 160 110 3.90 2.620 16.46 0 1 4 4 1970
Mazda RX4 Wag 21.0 6 160 110 3.90 2.875 17.02 0 1 4 4 1970
Datsun 710 22.8 4 108 93 3.85 2.320 18.61 1 1 4 1 1973
Hornet 4 Drive 21.4 6 258 110 3.08 3.215 19.44 1 0 3 1 1974
Hornet Sportabout 18.7 8 360 175 3.15 3.440 17.02 0 0 3 2 1977
Valiant 18.1 6 225 105 2.76 3.460 20.22 1 0 3 1 1962
```

If you would like to add a new row to the data, use the `rbind()` function, as follows:

```
> v <- data.frame(21.4, 4, 121.0, 109, 4.11, 2.780, 18.60, 1,
1, 4, 2, 1966)
> names(v) <- c("mpg", "cyl", "disp", "hp", "drat", "wt",
"qsec", "vs", "am", "gear", "carb", "year")
> rownames(v) <- c("Volvo 142E")
> rbind(new_mtcars, v)
mpg cyl disp hp drat wt qsec vs am gear carb year
```